Swiss Public Sector Job Market Analysis

Analysis - CIP Project Group 106

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1 Introduction

This project analyzes job advertisements within the Swiss public sector to examine how job requirements and working conditions vary across regions, occupations, and organisation types (cantons, municipalities, hospitals, etc.). It demonstrates how web-scraped data can be gathered, cleaned, and analysed within a complete data-science workflow to answer concrete research questions through visualisation and statistical analysis. The primary data source is publicjobs.ch, a dedicated job platform for the Swiss public sector, scraped via Selenium and BeautifulSoup. To contextualise the observed workload distribution against the broader Swiss labour market, the dataset is enriched with official employment statistics from the Swiss Federal Statistical Office (BFS), specifically the *Beschäftigungsstatistik* (BESTA) retrieved via the PX-Web API.

The analysis is guided by three research questions: **(RQ1)** How does Swiss public-sector labour demand distribute geographically and by occupational class? **(RQ2)** Are job postings statistically distinguishable from the BFS workforce benchmark in terms of full-time / part-time split? **(RQ3)** Which fringe benefits are most commonly advertised, and how do they vary between part-time and full-time roles and across cantons?

2 Methods

Data sources. Two complementary sources are used. First, 1'056 job listings are collected from publicjobs.ch (March 2026 snapshot, 73 pages × 20 listings) via a Selenium-based dynamic scraper that handles JavaScript rendering, cookie banners, and pagination, with randomised polite delays. For each listing, a second scraping pass visits the detail page in parallel (thread pool) to extract advertised employee benefits. Second, the BFS BESTA reference dataset (table `px-x-0602000000_101`, "Beschäftigte nach Wirtschaftsabteilung, Beschäftigungsgrad und Geschlecht") is retrieved via the PX-Web API and parsed from JSON-stat2 format into a tidy DataFrame. Compliance with `robots.txt` was verified before scraping.

Cleansing. Posting dates are parsed from `DD.MM.YYYY`, FTE columns are converted to numeric and reversed ranges (e.g. `100 - 60`) are corrected, duplicates and rows with missing critical fields are dropped. Canton URL slugs are normalised to standard German canton names. Each listing is enriched with (i) a heuristic seniority class (Einstieg / Fachkraft / Projektleitung / Stellvertretung / Führung) derived from title keywords, and (ii) an ISCO-08 major group derived from the publicjobs.ch occupation field via an explicit mapping table. Because publicjobs.ch truncates long category names at 33 characters, the ISCO mapping is augmented with the truncated forms so the join is complete (1'056 / 1'056 listings classified). Benefit chips are merged in via the listing's `detail_url` and one-hot encoded.

Analysis. RQ1 is addressed descriptively with a canton-level choropleth, a seniority bar chart, and an ISCO bar chart. RQ2 combines an exploratory boxplot of FTE by seniority and a stacked-bar chart of part-time / full-time share by canton with a confirmatory χ^2 test of independence comparing the 2026Q1 listings against the 2025Q1 BFS BESTA benchmark. RQ3 ranks the top 10 benefits by overall frequency, splits the rankings between part-time and full-time postings, and visualises geographic variation with a canton × benefit heatmap. The full reproducible code, including the scraper and the cleansing pipeline, is available in the project repository on GitHub and in the HTML render of this document; in the PDF report below, code chunks are suppressed for readability and only the resulting figures and discussion are shown.

3 Results and Discussion

3.1 RQ1: Geographic Distribution and Occupational Composition

The choropleth map (Figure 1) shows that the canton of **Zürich** has by far the highest number of job offers, accounting for 534 of 1'056 listings (50.6%) - more than all other cantons combined. **Luzern** follows with 98 postings (9.3%), while **Bern** and **Aargau** are tied at 88 postings each (8.3%). The remaining cantons are spread much more thinly, with a median of about 23 listings per canton and a long tail of smaller cantons contributing only a handful of openings. Even relative to Zürich's large population (~18% of Switzerland's residents), its share of public-sector listings (50.6%) corresponds to a per-capita over-representation of roughly **2.8×**, indicating that the imbalance is not merely a population effect but a real concentration of public-sector recruiting activity (and/or platform coverage) in the Zürich region.

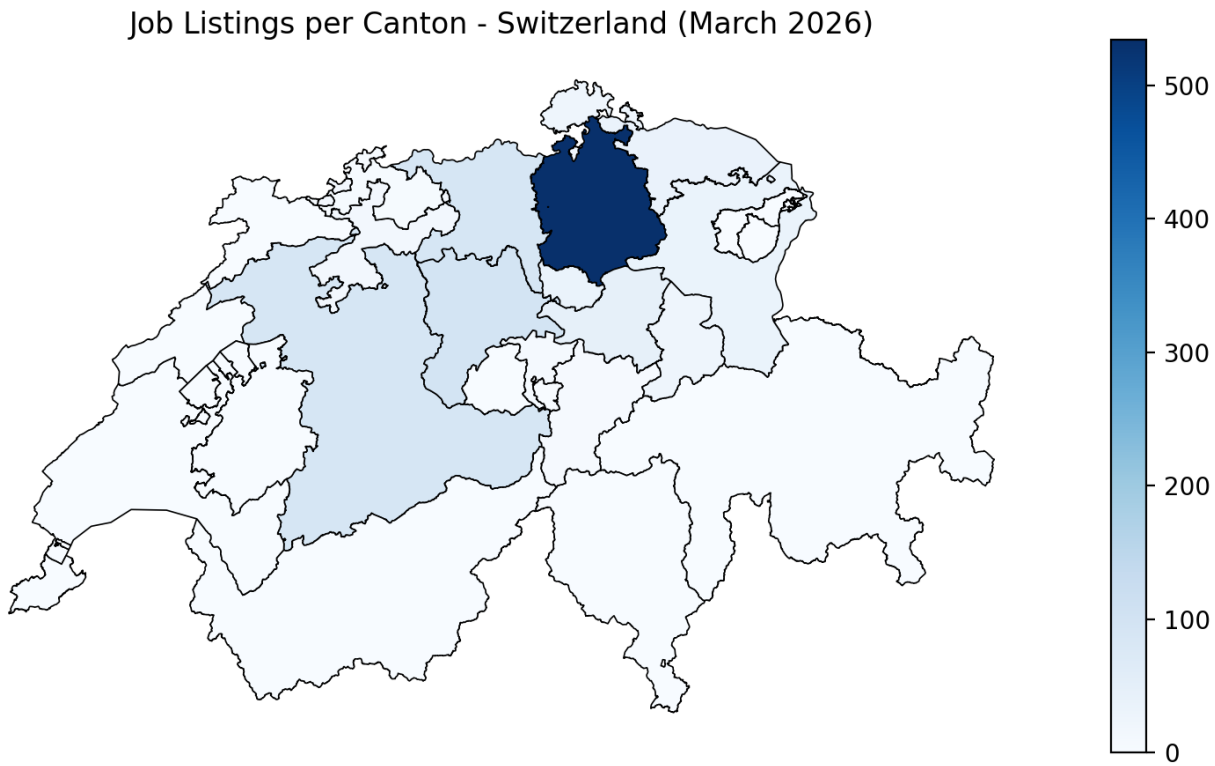

Figure 1: Geographic distribution of public sector job listings across Swiss cantons (March 2026).

We further categorised the job offers by seniority (Figure 2) and by ISCO-08 major group (Figure 3). The overwhelming majority of positions (84%) are classified as **Fachkraft** - qualified professionals holding a completed vocational training or academic degree but without an explicit leadership title. Explicit **Führung** (management) titles account for roughly 6% of postings, **Einstieg** roles (internships, trainees, juniors) for about 5%, **Projektleitung** (project leads) for around 3%, and **Stellvertretung** (deputies) for just over 1%. The ISCO classification tells a complementary story: **Akademische Berufe** dominate at 71% of listings (747 of 1'056), driven by healthcare, education, and public-administration roles, followed by **Bürokräfte und verwandte Berufe** at 13% (139), with service occupations (6%), craft workers (5%), elementary occupations (5%), and explicit leadership roles (1%) making up the remainder. Even though only 6% of titles carry an explicit Führung label, the ISCO mapping reveals that more than two-thirds of postings require academic-level qualifications - an asymmetry indicating that public-sector advertisers brand most openings on the role's professional content rather than on its place in a management hierarchy.

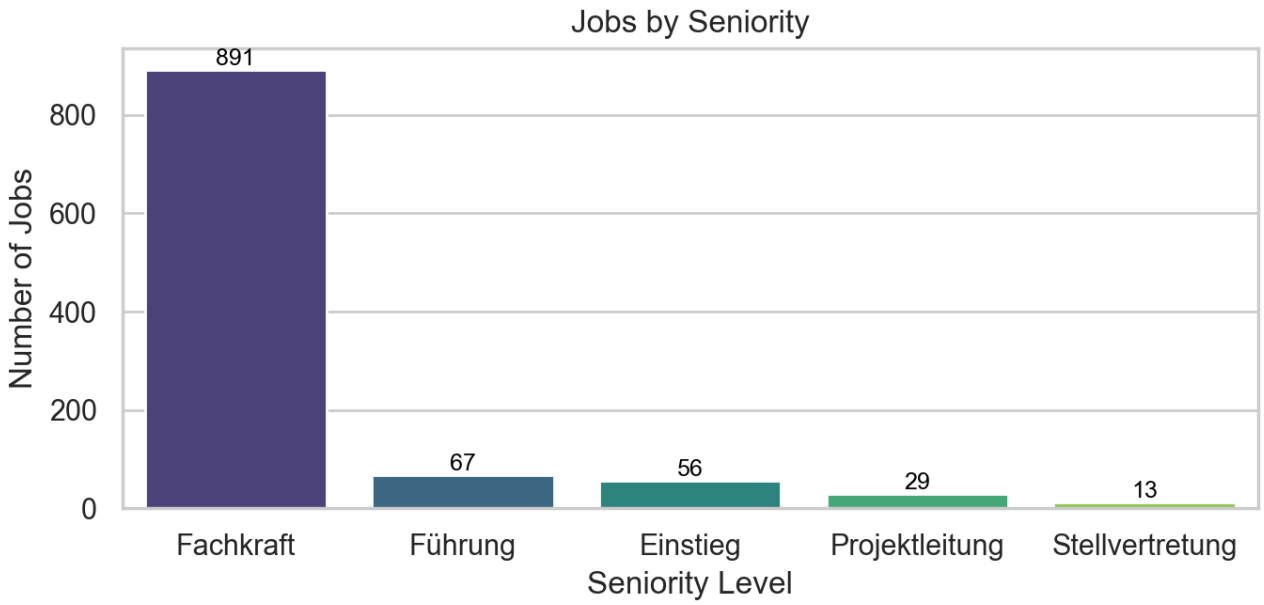

Figure 2: Job listings by seniority level.

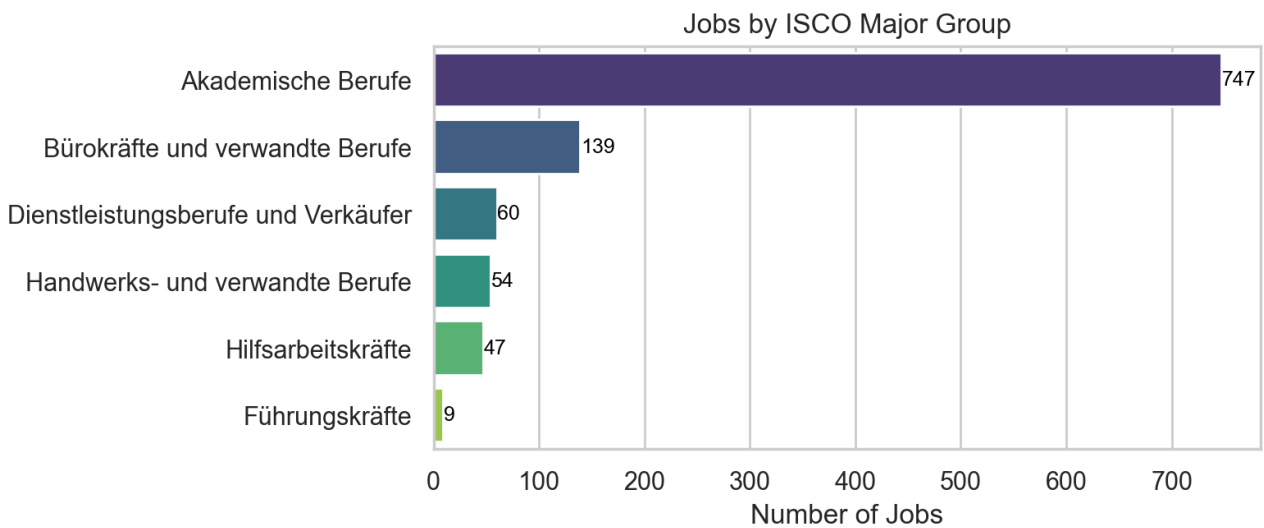

Figure 3: Job listings by ISCO-08 major group.

3.2 RQ2: Workload Flexibility - Public Sector Jobs vs. Swiss Labour Market

For 2026Q1, **59% of advertised positions are part-time (626 of 1'056)** and **41% are full-time (430 of 1'056)**. The official BFS BESTA reference for 2025Q1 shows the exact opposite split: part-time roles account for 41% (2'262'815 of 5'511'284) and full-time for 59% (3'248'469 of 5'511'284) of total employment in Switzerland (Figure 4). A χ^2 test of independence (H_0: identical part-time / full-time distribution in postings and BFS reference; H_1: distributions differ) on the raw counts rejects H_0 at any conventional level ($\chi^2 = 144.11$, $df = 1$, $p \approx 3.36 \times 10^{-33}$). Because the BFS sample size alone (≈ 5.5 million) makes any non-trivial difference statistically significant, we additionally compute Cohen's h on the two part-time proportions, a sample-size-independent effect-size measure: $h \approx 0.37$, which falls in Cohen's **small-to-medium** range and confirms that the gap is not merely an artefact of statistical power. The advertised jobs in this sample are therefore not only significantly but also substantively skewed toward part-time arrangements relative to the overall workforce - a pattern most plausibly driven by healthcare, social services, and education, where part-time work is structurally more common.

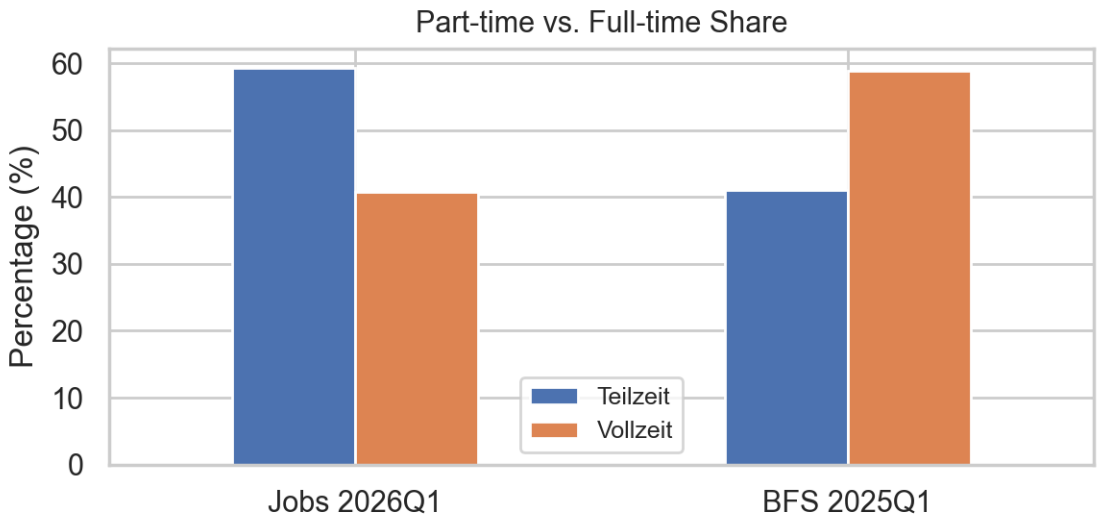

Figure 4: Workload share: publicjobs.ch listings (2026Q1) vs. BFS BESTA employment benchmark (2025Q1).

The share of part-time positions also varies markedly across cantons (Figure 5). Among cantons with a meaningful number of listings, **Luzern** stands out at the top with roughly 80% of its listings advertised as part-time, closely followed by **Schwyz** (74%) and **Zürich** (63%). **Aargau** sits near the middle of the distribution at around 57%, while **Bern** shows a clearly lower share of about 40%. At the opposite end of the spectrum, **St. Gallen** advertises only 11% of its postings as part-time, and **Solothurn** shows no part-time listings at all (0 of 15). These cross-canton differences are largely consistent with the underlying occupational mix of each canton's snapshot: Luzern, Schwyz and Zürich are dominated by healthcare, education and social-services postings, where part-time is the structural norm, whereas the few listings from Solothurn and St. Gallen in this sample are concentrated in security, technical and craft roles that are traditionally advertised full-time. The seniority breakdown of FTE (visible in the box-plot of average FTE per seniority level, see HTML render) further confirms that **Fachkraft** roles offer by far the widest distribution of working hours, while project-leadership and deputy positions are almost never advertised below 60% FTE - meaning workload flexibility is structurally limited at the management level and concentrated in the qualified-professional segment.

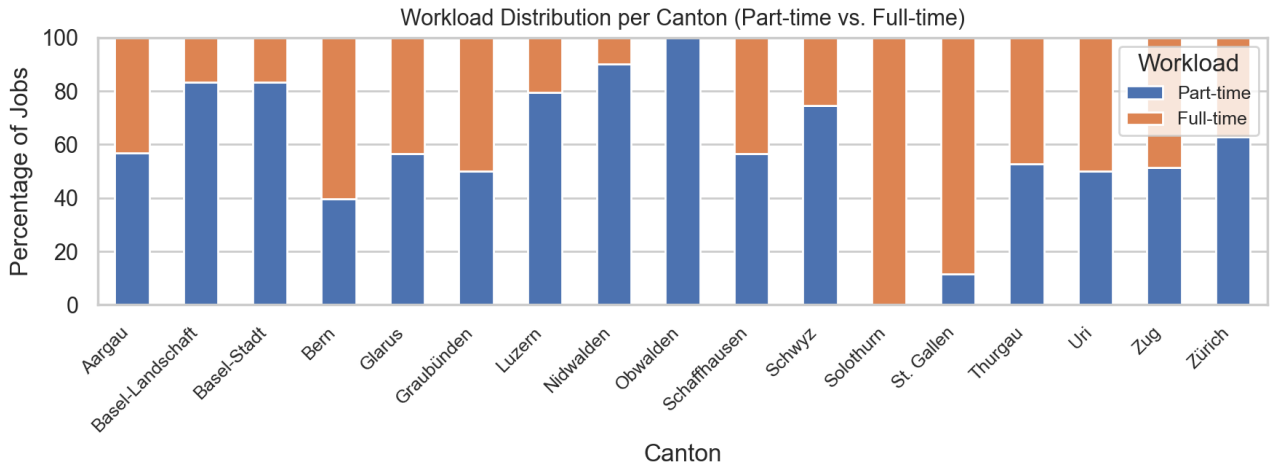

Figure 5: Part-time vs. Full-time share by canton.

3.3 RQ3: Benefits and Attractiveness

The benefit analysis (Figure 6) shows that Swiss public-sector employers predominantly advertise non-monetary fringe benefits. **Mitarbeitendenförderung** (staff development support, 48% of listings) and **Gute Vorsorgeleistungen** (strong pension provisions, 46%) are the most frequently offered benefits, reflecting the sector's traditionally strong focus on employment security and long-term stability. **Gleitzeit** (flexible working hours, 37%) and **ÖV-Anbindung** (public-transport accessibility, 35%) also feature prominently, consistent with the predominantly urban location of the listed positions.

The distribution of these benefits is uneven between part-time and full-time job offerings. In general, benefits are mentioned less frequently in part-time postings than in full-time ones. The gap is particularly pronounced for **Mitarbeitendenförderung** (66% full-time vs. 36% part-time), **Bezahlte Pausen** (paid breaks, 48% vs. 20%), and **Beitrag Aus- und Weiterbildung** (training contributions, 46% vs. 23%) - effectively a factor of roughly two. The notable exception is **Gute Vorsorgeleistungen**, which is offered at a nearly identical rate in both workload types (45% vs. 46%), in line with Swiss occupational-pension regulations that apply irrespective of contract size. Geographically (Figure 7), **Zürich** clearly provides the widest range of employee-support benefits in absolute terms - unsurprising given it hosts more than half of all listings - followed by a second tier of **Bern**, **Luzern**, and **Aargau**. The canton-level benefit profile largely mirrors listing volume rather than indicating structurally different benefit policies across regions. In summary, the availability of fringe benefits is not evenly distributed across cantons or contract types: full-time positions tend to advertise roughly twice as many top-tier benefits as part-time roles, with the important exception of pension provisions, which remain consistent across contract types and reinforce the sector's identity as a stable, long-term employer.

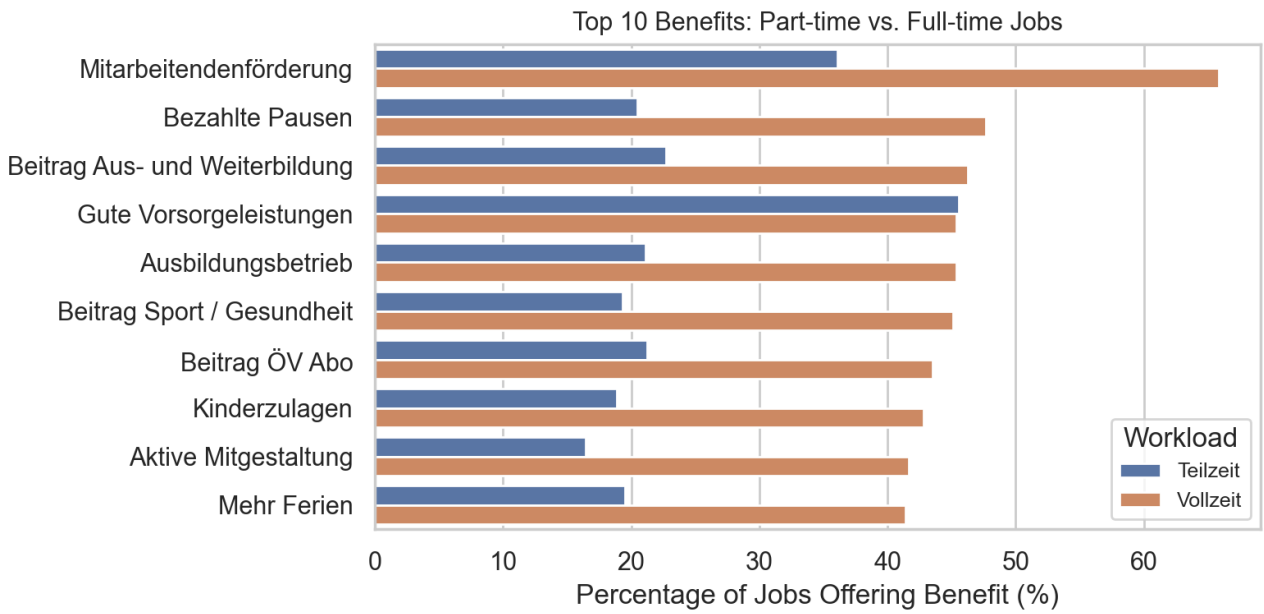

Figure 6: Top-10 benefits advertised in public-sector listings, split by part-time / full-time.

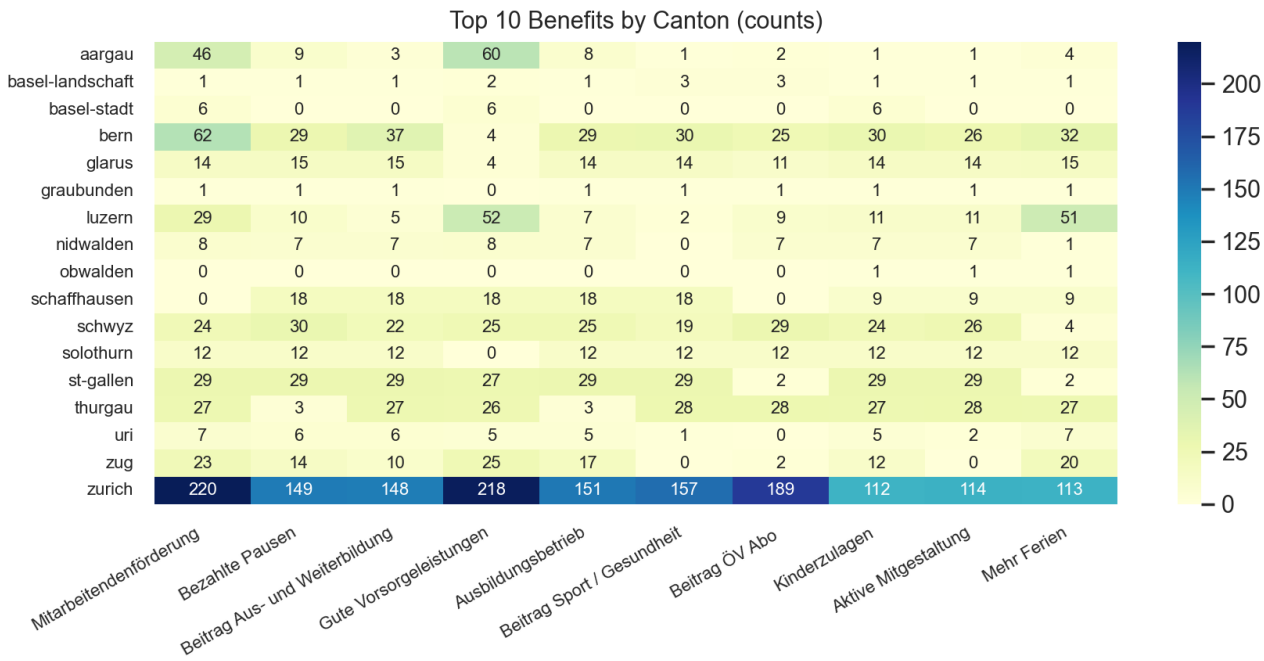

Figure 7: Top-10 benefits occurrence by canton (absolute counts).

4 Limitations and Future Work

Several limitations should be considered when interpreting these findings. First, the dataset represents a single cross-sectional snapshot of publicjobs.ch as of March 2026. Job posting volumes may exhibit seasonal patterns - for instance, public institutions may recruit more heavily at the start of an academic or fiscal year - and without repeated scraping at different points in time it is not possible to determine whether the observed distribution is representative of typical demand or reflects seasonal peaks. Whether such seasonality exists, and how large its effect might be, remains an open question.

Second, the platform covers the public sector exclusively. Findings do not generalize to private-sector employment, and any comparison with the broader Swiss labour market (e.g., against BFS BESTA figures) should be interpreted as a structural contrast rather than a direct equivalence.

Third, canton assignment is inferred from the URL slug of the job listing rather than from a verified location field. In cases where an employer's registered address differs from the actual workplace, or where positions are advertised cross-canton, this approach may produce misclassifications.

Fourth, the benefit extraction relies on successfully rendering each job's detail page. Any listings for which the detail page could not be loaded would have no benefit data, potentially biasing the benefit frequency estimates.

Fifth, the dataset relies on a single source platform. publicjobs.ch is one of several aggregators used by Swiss public-sector employers; institutions that recruit exclusively through their own portals, regional aggregators, or LinkedIn are systematically under-represented. The findings should therefore be read as reflecting the publicjobs.ch listing population rather than the entirety of the Swiss public-sector labour market.

For future work, periodic re-scraping at regular intervals would enable time-series analysis and allow seasonal effects to be quantified and controlled for. Extending the scraper to comparable platforms (e.g., jobs.ch, ostjob.ch, individual cantonal portals) would mitigate the single-platform bias and enable public-private sector comparisons. Applying NLP to the free-text job descriptions could yield richer features - such as required qualifications, language requirements, or contract duration - that are not available in the structured metadata fields.

5 Conclusion

This project demonstrates that publicly available job advertisement data, collected via automated web scraping, can yield structured insights into the Swiss public sector labour market. Across 1,056 listings from publicjobs.ch, three consistent patterns emerge. Geographic demand is heavily concentrated in Zürich, with canton-level disparities likely reflecting population size and institutional density rather than deliberate recruitment policy. The workload analysis reveals that public-sector advertisers disproportionately offer part-time positions relative to the BFS employment benchmark - a structural characteristic likely driven by healthcare and education, where flexible arrangements are common practice. Benefit profiling confirms a sector-wide emphasis on long-term employment security, flexible working hours, and commuting support, with relatively little variation across cantons or contract types.

From a methodological perspective, the project illustrates both the potential and the constraints of scraping-based data collection: the dataset is broad and timely, but its snapshot nature limits causal or longitudinal inference. Future work building on this foundation - particularly repeated scraping cycles and cross-platform comparison - would substantially increase the analytical depth achievable with this approach.

6 Appendix: Use of GenAI

Throughout this project, Claude was used as a supporting tool at various stages of the workflow. The group developed the project concept, research questions, and analytical approach independently. During implementation, Claude assisted iteratively with code development - particularly for the dynamic web scraper, where the team worked through the site's HTML structure, DOM selectors, and Selenium-based interaction logic (including pagination and cookie handling) step by step, validating each component against the live website. Claude also provided support during statistical analysis and visualisation implementation, as well as in polishing written documentation. All outputs were reviewed and validated by the group before inclusion in the final submission.